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# Cross Points and Markers Influence Analysis

## Automated Run Report

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## Run Configuration

Table 1: Runtime parameters loaded from the XML configuration file.

| Parameter                        | Value                                                                |
|----------------------------------|----------------------------------------------------------------------|
| Input source                     | excel [1 file(s)]                                                    |
| Output path                      | data/output/ALL_VISITS_output/P74_#2015m01d14-P74_V2365_FreeProtocol |
| Detail output file               | cpm_ia_detail.csv                                                    |
| Aggregated output file           | cpm_ia_aggregated.csv                                                |
| Threshold $\theta$ (%)           | 300.0                                                                |
| Rise tolerance $\varepsilon$ (%) | 1.0                                                                  |
| Max descent window $n_{\max}$    | 1000                                                                 |

## Dataset Overview

Table 2: Input dataset dimensions after schema validation and ingestion (Module 02).

| Metric                                  | Count     |
|-----------------------------------------|-----------|
| Total rows ingested                     | 80        |
| Rows dropped (empty keys)               | 0         |
| Invalid points found / rows affected    | 0 / 0     |
| Total anchors detected                  | 1         |
| Unique visits                           | 1         |
| Unique anatomical points                | 10        |
| Unique markers across all visits        | 8         |
| (Visit, point) series analysed          | 10        |
| Positive (visit, point) series          | 1 (10.0%) |
| Descent windows truncated at $n_{\max}$ | 0 (0.0%)  |

## Algorithm Description

The CPM-IA component analyses whether excitation events detected in bioelectrical surface current signals systematically influence the signal values recorded at subsequent measurement markers. The pipeline processes long-format tabular measurements through eight sequential analytical modules (M03–M10), each producing typed, immutable outputs consumed by the next stage.

### Input Data

The component ingests a single long-format CSV file (UTF-8, comma-separated). Each row encodes one (visit, anatomical point, marker) measurement triplet. The required columns are:

- **Visit identifier** — unique alphanumeric code for the acquisition session.
- **Marker identifier** — unique code identifying the stimulus or measurement marker.
- **Anatomical point identifier** — code identifying the electrode or body-surface acquisition site.

- **Percentage variation** — raw signal change relative to a reference value, expressed as a floating-point percentage (%).

Additional visit-level metadata columns (e.g. acquisition date) are preserved in the detail output without modification.

### Configuration Parameters

Three parameters govern the analysis; all are loaded exclusively from the XML configuration file at run time and are never hard-coded:

- **Threshold  $\theta = 300.0$  %**: a marker is classified as a *positive excitation* when its percentage variation meets or exceeds  $\theta$  ( $\geq \theta$ , inclusive).
- **Rise tolerance  $\varepsilon = 1.0$  %**: the maximum increase permitted between two consecutive values inside the descent window; a rise strictly exceeding  $\varepsilon$  closes the window.
- **Maximum window length  $n_{\max} = 1000$** : the descent window is forcibly closed after  $n_{\max}$  markers even when no rise exceeding  $\varepsilon$  has occurred (truncated window).

### Processing Steps

**M03 — Marker Sequence Reconstruction.** For each visit independently, the canonical marker detection order is derived from the first-appearance order of marker identifiers in the input file’s rows for that visit. This per-visit temporal ordering is immutable for the entire run: no sorting, grouping, or reordering of markers is performed after this step. Each (visit, anatomical point) pair is represented as an ordered list of percentage variation values aligned to its visit’s detection order; positions where a marker belongs to the visit protocol but was not measured at a specific point receive a NaN sentinel value.

**M04 — All-Positive Trigger Detection.** Each (visit, point) sequence is scanned in temporal detection order. *All* markers whose percentage variation *meets or exceeds*  $\theta$  ( $\geq \theta$ , inclusive) are designated *excitation anchors*. The full list of anchors is recorded as **positives\_count** per (visit, point) series. Series in which no marker meets  $\theta$  are flagged as “no cross-marker effect” and are excluded from all subsequent analytical modules (M06–M09).

**M05 — Positive Census.** An anatomical point is classified as *positive* if at least one of its visits produced an anchor. The census counts positive and total unique anatomical points across the dataset.

**M06 — Descent Window Determination.** For each detected anchor, starting from that anchor (step  $i = 0$ ), the window extends to successive markers as long as the signal does not rise by more than  $\varepsilon$  between consecutive steps. The window closes when any of three conditions is met: (1) a rise  $v_i - v_{i-1} > \varepsilon$  is encountered (**window\_close\_reason = rise**); (2) the next anchor position is reached (**window\_close\_reason = next\_positive**); (3)  $n_{\max}$  markers have been collected (**window\_close\_reason = n\_max**). If the sequence is exhausted without any of the above, the reason is **end**. The anchor is always included as the first element of the window. The **window\_close\_reason** field is recorded for each window in the detail output.

**M07 — Descent Slope (OLS Linear Regression).** An ordinary least-squares (OLS) linear regression is fitted to the  $L$  descent window values  $y_0, y_1, \dots, y_{L-1}$  at integer step positions  $x_i = i$ , where  $x_0 = 0$  corresponds to the anchor. The closed-form slope estimator is:

$$\hat{\beta}_1 = \frac{L \sum_{i=0}^{L-1} i y_i - \sum_{i=0}^{L-1} i \cdot \sum_{i=0}^{L-1} y_i}{L \sum_{i=0}^{L-1} i^2 - \left( \sum_{i=0}^{L-1} i \right)^2}$$

The slope  $\hat{\beta}_1$  (in % step<sup>-1</sup>) quantifies the average linear rate of change of the signal across the descent window. A window of length  $L = 1$  yields an undefined slope, reported as missing.

**M08 — Descent Descriptors.** Three scalar shape descriptors are computed for each positive (visit, point) series:

- **Descent depth** =  $v_{\text{anchor}} - \min(v_0, \dots, v_{L-1})$ , in %: total signal drop from the anchor to the lowest point in the window.
- **Descent length** =  $L$ : number of markers in the window, including the anchor.
- **Mean per-step decrease** =  $\text{depth} / (L - 1)$ , in % step<sup>-1</sup>: average drop per marker step, normalising depth by window duration. Undefined for  $L = 1$ .

**M09 — Variance Indicator.** Sample variance (denominator  $n - 1$ , i.e. `ddof` = 1) is computed separately on two signal segments:

- **Before-segment**: all sequence values at positions before the anchor index, after removing NaN entries.
- **After-segment**: the descent window values  $v_0, \dots, v_{L-1}$ , after removing NaN entries.

A segment with fewer than two clean values yields an undefined variance (reported as missing). The **variance ratio** =  $\sigma_{\text{after}}^2 / \sigma_{\text{before}}^2$  is computed when both variances are defined and  $\sigma_{\text{before}}^2 > 0$ . A ratio  $> 1$  indicates that signal variability *increases* after the excitation onset; a ratio  $< 1$  indicates it *decreases*.

**M10 — Output Consolidation.** Results from M04–M09 are assembled into two tidy output files:

- **Detail file** (`cpm_ia_detail.csv`): one row per (visit, anatomical point, *anchor*) triplet. Each anchor detected by M04 produces a separate row with columns `anchor_rank`, `anchor_marker`, `anchor_value`, `anchor_index`, and `window_close_reason`. (Visit, point) pairs with no cross-marker effect produce a single row with `anchor_marker` = None.
- **Aggregated file** (`cpm_ia_aggregated.csv`): one row per anatomical point, containing visit counts, positive prevalence, `median_positives_count`, and median descent descriptors computed over *all anchors* of that point.

## Output Artefacts

The pipeline produces three artefacts in the output directory `data/output/ALL_VISITS_output/P74_#2015m01d`

1. The per-(visit, point) detail CSV (M10).
2. The per-point aggregation CSV (M10).

3. This automated PDF run report (M11), generated from a filled L<sup>A</sup>T<sub>E</sub>X template and compiled with pdf<sub>l</sub>at<sub>e</sub>x in two passes to resolve table-of-contents page numbers.

No output file is ever silently overwritten: the pipeline raises an error if a target file already exists (guard G-10).

## Per-Point Results

Of the **10** (visit, point) series analysed, **1** (**10.0** %) yielded at least one excitation event above the threshold  $\theta = 300.0\%$ . **1** of **10** unique anatomical points have at least one positive visit. All tables in this section are sorted alphabetically by point identifier.

### Visit Counts and Positive Prevalence

A visit is *positive* for a given anatomical point when at least one marker in that (visit, point) series has a percentage variation meeting or exceeding  $\theta = 300.0\%$  ( $\geq \theta$ ) in temporal detection order (Module M04). All other visits are classified as “no cross-marker effect” for that point.

The **positive prevalence** is the proportion of total visits that are positive:

$$\text{Prevalence (\%)} = \frac{\text{Positive Visits}}{\text{Total Visits}} \times 100$$

A high prevalence indicates that the excitation event is consistently reproduced at a given point across visits; a low prevalence indicates that the event is rare or confined to specific sessions.

| Anatomical Point | Total Visits | Positive Visits | Prevalence (%) |
|------------------|--------------|-----------------|----------------|
| LF - 5 - e       | 1            | 1               | 100.0          |
| LF - 5 - i       | 1            | 0               | 0.0            |
| LH - 3 - i       | 1            | 0               | 0.0            |
| LH - 4 - e       | 1            | 0               | 0.0            |
| LH - 5 - e       | 1            | 0               | 0.0            |
| RF - 5 - e       | 1            | 0               | 0.0            |
| RH - 2 - i       | 1            | 0               | 0.0            |
| RH - 3 - i       | 1            | 0               | 0.0            |
| RH - 4 - e       | 1            | 0               | 0.0            |
| RH - 5 - e       | 1            | 0               | 0.0            |

### Most Frequent Trigger Markers by Anatomical Point

For each anatomical point, the table shows which markers most frequently acted as an *excitation anchor* — any marker in temporal detection order (Module M03) whose percentage variation met or exceeded  $\theta = 300.0\%$  (Module M04) — across positive visits.

Only markers that acted as anchor in at least **3.5** % of the positive visits for that point are listed. Markers with a lower share are scattered across sessions with no single dominant trigger and are omitted for readability. All listed marker values met or exceeded  $\theta = 300.0\% > 100\%$  in the visits where they acted as anchor. Anatomical points whose most frequent anchor marker appears in fewer than 3.5 % of positive visits do not appear in this table (no single marker is dominant at those points).

### How to read the columns:

- **Trigger Marker:** identifier of an anchor marker that met or exceeded  $\theta$  for that point.

- $n$ : number of positive visits for that point in which this marker was the anchor.
- **Share of Pos. Visits (%)**:  $n$  expressed as a percentage of the total positive visits for that point. Values near 100 % indicate a dominant, reproducible trigger; lower values indicate concurrent triggers across sessions (e.g. due to protocol or biological variability).

Rows are sorted by anatomical point (alphabetical) and, within each point, by descending  $n$ .

| Anatomical Point | Trigger Marker | $n$ | Share (%) |
|------------------|----------------|-----|-----------|
| LF - 5 - e       | IFN-beta       | 1   | 100.0     |

### Descent Shape Medians (Positive Visits Only)

All values in this table are medians computed over *all anchors* of positive visits — visits in which at least one excitation anchor was detected for that point. Each anchor contributes one row to the detail file and one value to each median. Points with no positive visits receive a dash (—) for all descriptor columns. Medians are computed on the raw per-anchor values; no smoothing or outlier removal is applied.

#### How to read and interpret the columns:

- **Med. Slope** ( $\% \text{ step}^{-1}$ ): median OLS linear regression slope across the descent window (Module M07). A negative value indicates a consistent downward trend in the post-excitation signal; a slope near zero indicates a flat or irregular window. The slope is the rate-of-change estimator that minimises the sum of squared vertical residuals between the fitted line and the window values. Undefined for single-marker windows (—).
- **Med. Depth** (%): median total signal drop from the anchor value to the minimum value in the descent window,  $v_{\text{anchor}} - \min_{\text{window}}$  (Module M08). Larger depth indicates a more pronounced excitation response; a depth near zero means the signal barely decreases after the anchor.
- **Med. Length** (marker steps): median number of markers in the descent window, including the anchor (Module M08). A longer window indicates a sustained post-excitation decline before a natural rise exceeding  $\varepsilon = 1.0\%$  is observed or before the  $n_{\text{max}} = 1000$  cap is reached. A length of 1 means the signal rose immediately at the next marker after the anchor.
- **Med. Step Decr.** ( $\% \text{ step}^{-1}$ ): median mean per-step decrease, computed as  $\text{depth} / (\text{length} - 1)$  (Module M08). This metric normalises the total drop by the window duration, providing a rate-of-descent estimate that is comparable across windows of different lengths. Undefined for single-marker windows (—).

**Interpretation guidance.** Anatomical points with high positive prevalence (Section 4.1), large median depth, and a well-defined negative slope show a strong and consistent post-excitation descent across visits. Points with low prevalence but large depth when positive are intermittently reactive. A high median length combined with a moderate slope may indicate a gradual, sustained descent rather than a sharp drop. The variance ratio (available in the detail CSV, column `variance_ratio`) complements these descriptors by quantifying whether signal dispersion increases or decreases after the excitation onset.

| Anatomical Point | Med. Slope | Med. Depth | Med. Length | Med. Step | Decr. |
|------------------|------------|------------|-------------|-----------|-------|
| LF - 5 - e       | -76.36     | 209.09     | 4.0         |           | 69.70 |
| LF - 5 - i       | —          | —          | —           |           | —     |
| LH - 3 - i       | —          | —          | —           |           | —     |
| LH - 4 - e       | —          | —          | —           |           | —     |
| LH - 5 - e       | —          | —          | —           |           | —     |
| RF - 5 - e       | —          | —          | —           |           | —     |
| RH - 2 - i       | —          | —          | —           |           | —     |
| RH - 3 - i       | —          | —          | —           |           | —     |
| RH - 4 - e       | —          | —          | —           |           | —     |
| RH - 5 - e       | —          | —          | —           |           | —     |